

Datathon Assignment Outline and Marking Guide (2026): Insurance Pricing Promises & Transparency

This assignment is developed in partnership with Finity Consulting and grounded in two real ASIC regulatory findings into insurer pricing conduct (not just another assignment)! You will step into the role of a general insurer's pricing team and design a rating structure that satisfies both your Chief Actuary and a regulator reviewing your work.

Summary

1. Objectives:

- a) Expanding your learning experience by working on real industry challenges with support from peers, academics, and industry practitioners
- b) Providing opportunities for you to develop skills (e.g. teamwork, communication, problem-solving skills) required for success in your future career by solving complex challenges in a collaborative setting
- c) Empowering you to solve real-life problems and get recognised by industry partners
- d) Supporting you to generate evidence on your creative solutions to real-world problems and employability skills for your professional profile to impress future employers

2. Tasks:

- a) In this project, you and your classmates will form a team to create feasible solutions to a real industry challenge provided by industry partners.

3. Assignment Format: Group Assignment

- a) Team size: 4-5 students
- b) Team formation method: self-select (i.e. you choose your own team)
- c) Team members: you can form a team with students from or outside of your tutorial. However, undergraduate (postgraduate) students are expected to form a team with other undergraduate (postgraduate) students only. Note as long as you all enrolled in the same course code (ACTL4305 or ACTL5305), you are allowed to form as one group.

4. Marking: 30 Marks (30% of the course total mark – note this is the nominal marks. This assignment will have spillover effect on the final exam, as final exam questions will be written partially based on the context of the Assignment.)

- 12 Marks: Progress Milestone Submission (Formative assessment)

Marks will be awarded based on participation and completion of the milestone task.

Milestone feedback will be provided to support the development of your final solution.

- 18 Marks: Final Solution Submission (Summative assessment)

Marks will be awarded according to the marking rubric attached in this document.

5. Deadlines:

- a) Team Formation: Thursday, 11:59pm 1 October (Sydney Time)
- b) Progress Milestone Submission: 13 Oct (group submission), 15 Oct (peer voting), and 25 Oct (individual reflection), all at 11:59pm (Sydney Time)
- c) Final Solution Submission: Sunday, 11:59pm 8 November (Sydney Time)

Industry Datathon Challenge: Assignment Details

"Firms have now resolved that university results are not necessarily a reliable means of identifying good new talent ... firms need (graduates who can) ... understand the needs of clients and applying creativity to identify holistic solutions."

- The Australian Financial Review, 22 July 2020

1. Overview

The project aims to enhance your career-focused learning experience by bringing in real-world scenarios and real industry challenges into the classroom, creating a safe space for you to explore, collaborate and make changes.

The assignment is intended to promote problem-based learning (PBL) in which you learn about a subject by working in teams to solve real-life problems. It is also intended to develop your skills in research and problem-solving, your ability to present your ideas concisely and coherently, and to work in groups.

Solving real-life problems is an inherently complex and messy process, but such a process also offers plenty of learning opportunities. You will learn about collaboration, working through problems persistently, seeking creative solutions, and being comfortable with changing paths where necessary.

Your experience in this project will be helpful in your transition into the professional environment – you will be prepared to leverage your existing knowledge and skills, while at the same time identify and act on knowledge and skill gaps, to respond to new challenges and seize emerging opportunities coming your way.

2. Team Formation

In this project, you and your classmates will form a team to create feasible solutions to a real industry challenge. You can form a team with students from or outside of your tutorial. However, undergraduate (postgraduate) students are expected to form a team with other undergraduate (postgraduate) students only. Students who fail to do so by the Team Formation Deadline will be randomly allocated to teams.

You are expected to form your own teams (team size: 4-5) via the “Assignment Group Self-selection” link on Moodle before the Team Formation Deadline - **Thursday, 1 October, 11:59pm (Sydney Time).**

3. Project Deliverables

I. Final Solution Submission (18 Marks)

- Submission Deadline: **Sunday, 11:59pm 8 November 2026 (Sydney Time)**
- Only ONE Final Solution Submission is required per team.
- Project information and requirements: see Assignment Brief on Moodle.
- Your Final Solution Submission should include:

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| <ul style="list-style-type: none">1) Submission Cover Page (available on Moodle) signed by ALL team members.2) Reproducible Code: R or Python code that runs end-to-end from the provided CSVs to your final priced books, with clear instructions on how to use it. |
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3) Report (PDF, no more than 10 pages, excluding the cover page, references and appendices). Tables and figures included in the main report count towards the 10-page limit.

4) Two output files: priced_2027.csv and priced_2028.csv, each with at minimum PolicyID, TechnicalPrice, FinalPrice, and the dollar value of each discount applied.

5) Video Presentation with slides (up to 10 minutes): Summarise your key findings and pricing design choices. It is not required that all members show up in the video.

- Marking criteria: See Assessment Criteria on the last page.

II. Progress Milestone Submission (12 Marks)

Objectives:

The Progress Milestone Submission is designed to support and facilitate your team's progress towards the delivery of your Final Solution.

Marking:

Marks for the Progress Milestone Submission will be awarded based on participation and completion of the milestone tasks.

This design aims to provide a safe environment for your team to explore different solution options and receive ongoing feedback from your peers, academics, and Finity Consulting before submitting your Final Solution.

Only one Group Milestone Submission is required per team for Task 1. The Individual Milestone Submission in Task 2 must be submitted separately by each team member.

Marking criteria:

As a formative assessment task, the Progress Milestone Submission is designed to encourage your team to work on your project on a regular basis and provide ongoing feedback. You don't need to get the best solution right away, what counts is that your team makes a sincere attempt and learn as much as possible from the feedback.

To receive full marks for this component (12 marks), the required group and individual milestone tasks must be completed. You are strongly encouraged to attempt the milestone tasks with your best effort to ensure that you make solid progress towards the final solution, to receive useful feedback before your final submission, and to utilise all the opportunities for your team and solution to get recognised by Finity Consulting.

To complete Task 1, your team needs to:

- 1) **Share:** Share your team's milestone submission in the designated [MS Teams](#) Milestone Submission Channel (to be opened in Teams) by 11:59 pm (mid-night) of the due date (Tuesday) 13 Oct, and
- 2) **Vote:** Select two milestone submissions in the designated forum that your team found most insightful from other student teams by 11:59pm of the due date (Thursday) 15 Oct.

Please note, detailed guidelines regarding the milestone submission and voting forum will be announced on [MS Teams](#).

To complete Task 2, each team member needs to: The proposal looks very good to me. The structure is clear and the roles are well defined. One suggestion is to allow some interaction

between panel members after each section, so that the evidence, product implications and member strategies are connected throughout the session.

Submit an individual reflection of up to 3 pages via the **Moodle Milestone Submission Tool**, by 11:59pm of the due date (Sunday) 25 Oct.

Progress Milestone Submission Tasks

Task	Milestone Tasks	Marks	Submission Channel	Milestone Due Date	Vote Due Date
1.	Group Milestone Submission (to be prepared together by the team): 1. Team Introduction + Progress Video (max 10 mins) 1) Introduce all team members. 2) Present your initial exploratory data analysis findings and planned modelling approach. 2. 1-page summary: Provide a concise summary of your key data quality findings and modelling strategies. <i>Note: this milestone submission is to help you make progress and receive feedback from both peers and the lecturer.</i>	9 marks in total, including 7 marks for the submission and 2 marks for voting	MS Teams Channel for Milestone Submissions (Can be seen by other teams)	11:59pm 13 Oct (Tuesday)	11:59pm 15 Oct (Thursday)
2.	Individual Milestone Submission (to be prepared by each team member): A document of up to 3 pages, consisting of: 1. Your contributions to the group assignment's milestone submission ($\approx$ 1 page) 2. Reflections on: (1) the key findings of your group assignment so far, and (2) the skills you have learned from this group assignment (up to 2 pages)	3 marks	Moodle Milestone Submission Tool	11:59pm 25 Oct (Sunday)	NA

4. Project Support

I. Resources:

Resources and supporting materials (e.g. Examples & 'How to' guide) are available on Moodle to help your team produce project deliverables and complete the milestone tasks.

II. Consultation and Advice:

If your team needs help or advice on your project, please feel free to book a time to talk to your Project Advisor – Fei Huang (Lecturer) via the online 1-1 consultation booking tool on Moodle.

Please feel free to use other consultation hours available on Moodle.

III. Opportunities to Engage with Finity Consulting:

- Industry Partner Guest Talk (Week 5, Monday): Finity Consulting will join us during our Workshop for a guest talk. The session is designed for Finity Consulting to share their experience and perspectives on the challenges in the assignment and answer your questions.
- Q&A channel.
- Milestone Submissions: top milestone submissions voted by students will be shared with Finity Consulting for review and feedback.
- Top Groups Invited Onsite Presentation. The event will be attended by industry professionals as well as academics from UNSW. It will be an invaluable networking opportunity with industry partners!

IV. Dealing Conflict Resolution - Optional Peer Review

If you experienced a significant unequal contribution in a team, there would be an option for your team to initiate a Peer Review Process for your group project. However, this should only be used as a last resort. You are strongly encouraged to negotiate a resolution yourselves within the team before initiating this formal review.

This is a completely optional process for conflict resolution - teams where all members made equal contributions to the project do NOT need to undertake this exercise.

The details of the Optional Peer Review Process is available on Moodle under this Assignment's section.

Assessment Criteria - Final Solution Submission

The Final Submission Package (including videos and reports) will be assessed holistically against the following rubrics.

Criteria	Below Expectations <50% (FL)	Meets Expectations 50% - 74% (PS - CR)	Exceeds Expectations ≥ 75% (DN - HD)
Application of relevant business knowledge and skills to identify a problem and provide a solution (5 marks)	- Does not apply relevant business knowledge sufficiently, accurately to identify or address problems. - Does not synthesise relevant knowledge or skills; or show their application to problem solving.	- Applies relevant business knowledge largely accurately (only minor errors) and sufficiently to identify and solve problems. - Provides some (incomplete) synthesis of relevant knowledge and skills, with application to problem solving.	- Insightfully and accurately applies relevant business knowledge innovatively to identify and solve problems. - Appropriately integrates and synthesises relevant knowledge and skills, with application to problem solving.
Development of a feasible solution process to a well-articulated problem (8 marks)	- Does not develop a valid solution process justified by analysis, theory and/or research. - Does not reference sources appropriately/accurately in reference list.	- Develops a plausible and realistic solution process, although it may contain some weaknesses or over-simplification, and/or limited justification. - Does not sufficiently consider or evaluate differing perspectives or acknowledge limitations and constraints of own solution. - References sources accurately in reference list. Referencing shows some research.	- Develops a creative and feasible solution process suitable for real-life application. The solution proposed is well-supported by analysis, evidence, theory, or research. - Considers and evaluates differing perspectives and alternative strategies (if appropriate) and acknowledges limitations and constraints of own solution. - References sources accurately in reference list. Referencing shows extensive research.
Communication (5 marks)	- The solution submission is poorly structured and unclear. - Does not engage the audience or have a sufficiently professional delivery (e.g. videos and reports). - Does not use tools/technologies effectively (e.g. videos, visual aids).	- The solution submission is clear and well structured. - Achieves some audience engagement through a sufficiently professional delivery (e.g. videos and reports). - Generally, uses tools/technologies effectively (e.g. videos, visual aids)	- The solution submission is professional, innovative and engaging, suitable to be shared with the audience in the industry. - Engages the audience effectively through a professional delivery (e.g. videos and reports). - Uses tools/technologies very effectively (e.g. videos, visual aids)